

Name: _____

ALGEBRAIC EXPRESSIONS, (FACTORISING EXPRESSIONS)

Date: _____

LO: I can factorise algebraic expressions by taking out common factors.

What is factorising?

Factorising is the reverse of expanding. We find a **common factor** and write the expression as a product of terms.

Strategy: Find the HCF of all terms (both numbers and letters), then divide each term by it.

Quick examples

●	$6x + 9 = 3(2x + 3)$	HCF is 3
●	$x^2 + 5x = x(x + 5)$	HCF is x
●	$4x^2 + 8x = 4x(x + 2)$	HCF is 4x
●	$6x + 9 = 6(x + 3)$	6 is not a factor of 9
●	$x^2 + 5x = x(x + 5x)$	divided only first term

Key idea

Find the highest common factor (HCF) of all terms. Write it outside the bracket. Divide each term by the HCF to get what goes inside.

$$4x^2 + 8x = 4x(x + 2)$$

HCF = 4x; divide each term by 4x

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - number HCF

Factorise $6x + 9$

HCF of 6 and 9 is 3

$$6x \div 3 = 2x; 9 \div 3 = 3$$

Write: $3(2x + 3)$

$$= 3(2x + 3)$$

Check by expanding: $3 \times 2x + 3 \times 3 = 6x + 9$

Example 2 - letter HCF

Factorise $x^2 + 5x$

Both terms contain x

$$x^2 \div x = x; 5x \div x = 5$$

Write: $x(x + 5)$

$$= x(x + 5)$$

Take out the lowest power of x from each term.

Example 3 - number and letter

Factorise $4x^2 + 8x$

HCF of 4 and 8 is 4; both have x:

$$\text{HCF} = 4x$$

$$4x^2 \div 4x = x; 8x \div 4x = 2$$

Write: $4x(x + 2)$

$$= 4x(x + 2)$$

Always take the highest common factor, including letters.

Example 4 - three terms

Factorise $6x^2 + 12x + 18$

HCF of 6, 12, 18 is 6

$$6x^2 \div 6 = x^2; 12x \div 6 = 2x; 18 \div 6 = 3$$

Write: $6(x^2 + 2x + 3)$

$$= 6(x^2 + 2x + 3)$$

When all coefficients share a common factor, take it out.

YOUR TURN – PRACTICE (deliberate practice)

1. Number common factor

Factorise each expression.

- a) $4x + 8$
- b) $6x + 15$
- c) $10x - 5$
- d) $12x + 18$

2. Letter common factor

Take out the highest power of the letter.

- a) $x^2 + 3x$
- b) $x^2 - 7x$
- c) $2x^2 + x$
- d) $5x^2 - 10x$

3. Number and letter HCF

Find the full HCF (number and letter).

- a) $6x^2 + 12x$
- b) $8x^2 - 4x$
- c) $9x^2 + 3x$
- d) $10x^3 + 15x^2$

4. Three-term expressions

Factorise fully.

- a) $3x^2 + 6x + 9$
- b) $5x^2 - 10x + 15$
- c) $4x^2 + 8x + 12$
- d) $2x^3 + 4x^2 + 6x$

5. Mixed factorising

Factorise each expression fully.

- a) $7x + 21$
- b) $x^2 + 4x$
- c) $12x^2 - 8x$
- d) $15x^2 + 10x + 5$
- e) $6x^3 - 9x^2$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $6x + 9 = 3(2x + 3)$ ☐

Correct answer: _____

Reason: _____

b) $x^2 + 5x = x(x + 5x)$ ☐

Correct answer: _____

Reason: _____

c) $4x^2 + 8x = 4x(x + 2)$ ☐

Correct answer: _____

Reason: _____

d) $6x + 9 = 6(x + 1.5)$ ☐

Correct answer: _____

Reason: _____

e) Factorising is the reverse of expanding ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Factorise $12x^2y + 18xy^2$ fully. Explain how you found the HCF of both the numbers and the letters.

Expression: _____

Simplified: _____

b) Show that $n^2 + n$ is always even for any integer n . (Hint: factorise first, then explain why the product must be even.)

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

